bigger set. However, the opposite is not allowed, i.e. a partition can't be split into 2 subsets.

Now imagine that there was a design discussion between your data-science team and your support engineering team, and one engineer fiercely stood up and exclaimed: “well, that’s easy! You keep an array (a dynamic array or a vector) for each subset”.

You don’t want to be that person, believe me, and one of the goals of this book is making sure you don’t find yourself in that situation. Because the next thing to happen, hopefully, is that somebody else points out how, by using arrays, understanding if two elements are in the same set could potentially require going over all the elements of all the subsets. Likewise, just understanding which subset an element belongs to, could require the same number of element checks, i.e. linear in the total number of elements.

This would be a real performance concern, and it seems obvious we should be able to do better.

The next idea in this brainstorming could involve adding a map from elements to subsets, together with the list of subsets above. This is a slightly better improvement for some operations, although, as we will see, it still forces operations like merging two sets to potentially require `O(n)` assignments.

Performance, however, is not the main concern with that design. Using two independent data structures is a terrible idea, because you will have to manually sync them every